OS ORANGE-1

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SRN – PES1UG24AM101

SEC – B

```
hemanth@saihemanth: ~
hemanth@saihemanth:~$ vim clash_server.c
hemanth@saihemanth:~$ gcc clash_server.c -pthread -o clash
hemanth@saihemanth:~$ ./clash
[PES1UG24AM101] Clash Royale Match Starting...
[PES1UG24AM101] Arena Capacity: 6 | Max Elixir: 12
[PES1UG24AM101] Spawner 1 deployed a troop (Cost: 3 Elixir).
[PES1UG24AM101] Spawner 3 deployed a troop (Cost: 3 Elixir).
[PES1UG24AM101] Spawner 1 deployed a troop (Cost: 4 Elixir).
[PES1UG24AM101] Spawner 3 deployed a troop (Cost: 3 Elixir).
[PES1UG24AM101] Spawner 2 deployed a troop (Cost: 4 Elixir).
[PES1UG24AM101] Spawner 1 deployed a troop (Cost: 2 Elixir).
[PES1UG24AM101] Spawner 3 deployed a troop (Cost: 4 Elixir).
[PES1UG24AM101] Spawner 2 deployed a troop (Cost: 4 Elixir).
[PES1UG24AM101] Spawner 1 deployed a troop (Cost: 2 Elixir).
[PES1UG24AM101] Spawner 3 deployed a troop (Cost: 4 Elixir).
[PES1UG24AM101] Spawner 2 deployed a troop (Cost: 4 Elixir).
[PES1UG24AM101] Spawner 3 deployed a troop (Cost: 4 Elixir).
[PES1UG24AM101] Spawner 2 deployed a troop (Cost: 4 Elixir).
[PES1UG24AM101] Spawner 3 deployed a troop (Cost: 1 Elixir).
[PES1UG24AM101] Spawner 2 deployed a troop (Cost: 3 Elixir).
[PES1UG24AM101] Spawner 1 deployed a troop (Cost: 3 Elixir).
[PES1UG24AM101] Spawner 3 deployed a troop (Cost: 2 Elixir).
[PES1UG24AM101] Spawner 2 deployed a troop (Cost: 4 Elixir).
^C

=== KING TOWER DESTROYED! ===
[PES1UG24AM101] MATCH OVER (SIGINT Caught).
[PES1UG24AM101] Total Troops Deployed: 21
Closing server connections...
hemanth@saihemanth:~$
```

TODO-1

```
// TODO 1: IPC - Create the Battle Log Pipe
// ------------------------------------------------------------

//
    if (pipe(battle_log_pipe) == -1) {
            perror("pipe");
            exit(1);
    }
```

TODO-2

```
// TODO 2: SYNCHRONIZATION - Initialize Sync Primitives
// ------------------------------------------------------------

    sem_init(&arena_empty_slots, 0, arena_capacity);
            pthread_mutex_init(&arena_mutex, NULL);
            pthread_mutex_init(&elixir_lock, NULL);
```

TODO-3

```
// ------------------------------------------------------------
// TODO 3: SIGNALS - Handle Match Over
// ------------------------------------------------------------

    signal(SIGINT, match_over_handler);


register_match();
```

TODO-4

```
// TODO 4: DEADLOCK PREVENTION - Spend Elixir Atomically
// ------------------------------------------------------------
bool spend_elixir(int cost) {
    bool deployed = false;

     pthread_mutex_lock(&elixir_lock);

        if (available_elixir >= cost) {
                available_elixir -= cost;
                deployed = true;
        }

        pthread_mutex_unlock(&elixir_lock);

    return deployed;
}
```

TODO-5A

```
    // TODO 5a: BOUNDED BUFFER (Producer) - Deploy troop into Arena
    //    - Wait for an empty Arena slot
    //    - Lock the Arena, place the troop, advance deploy_idx
    //      (use circular indexing!), then unlock
    // ------------------------------------------------------------

    sem_wait(&arena_empty_slots);

    pthread_mutex_lock(&arena_mutex);

    arena[deploy_idx] = id;
    deploy_idx = (deploy_idx + 1) % arena_capacity;

    pthread_mutex_unlock(&arena_mutex);
```

TODO-5B

```
            // TODO 5b: BOUNDED BUFFER - Free the Arena slot
            //    - The troop's lifespan (sleep) is over. Signal that
            //      an Arena slot is available again.
            // ------------------------------------------------------

            sem_post(&arena_empty_slots);

        } else {
            usleep(150000);
        }
    }
```